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Nexarag: Democratizing Reproducible Knowledge Graph Contexts for LLM Research

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Software

- Review
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Summary

Large language models (LLMs) are widely used in research workflows but struggle with hallucinations, short context windows, and weak reproducibility in literature reviews (Huang et al., 2025; Ji et al., 2023). Nexarag is a modular, open-source platform that lets researchers curate, visualize, and share custom knowledge graphs (KGs) from academic sources stored in Neo4j (Neo4j Inc., 2024). Through native support for the Model Context Protocol (MCP), any MCP-compatible LLM can access these curated KGs for controllable, reproducible context injection (Anthropic, 2024; Model Context Protocol Contributors, 2024)—including fully private, air-gapped deployments via containers (Boettiger, 2015)—so teams can explore literature more deeply and transparently. Nexarag provides interactive graph/semantic visualizations using Cytoscape.js and D3 (Bostock et al., 2011; Franz et al., 2023).

Statement of need

Traditional retrieval-augmented generation (RAG) systems rely primarily on embedding-based similarity (Guu et al., 2020; Lewis et al., 2020; Reimers & Gurevych, 2019), which often misses long-range semantic structure and relationships across documents, especially in long-context, multi-document settings (Gao et al., 2023; Wang et al., 2024). This weakens controllability, auditability, and reproducibility for complex research tasks. Knowledge graphs provide a structured and interpretable alternative by modeling entities and relations explicitly (Reinanda et al., 2020). However, existing KG-powered tooling is either proprietary and expensive or technically demanding to deploy and maintain. Nexarag fills this gap with a researcher-friendly platform that automates KG creation from academic inputs, supports semantic exploration, and standardizes LLM access via MCP (Anthropic, 2024), enabling reproducible literature synthesis across local and cloud settings.

State of the field

RAG improves access to external knowledge and has become a standard strategy for knowledge-intensive NLP (Gao et al., 2023; Guu et al., 2020; Lewis et al., 2020), yet it struggles with long contexts and multi-step reasoning when similarity search is the only primitive (Wang et al., 2024). KGs address this by enabling path-based queries and explicit relation reasoning while preserving transparency and updatability (Reinanda et al., 2020; Sahlab et al., 2022; Xu et al., 2024). Nexarag's contribution is to operationalize these advantages in a package that researchers can run locally, share with collaborators, and connect to a wide range of LLM hosts through MCP (Anthropic, 2024).

Software overview

Core capabilities. Nexarag provides: (i) automated KG construction from BibTeX, paper lists, search queries, and citation expansion (Semantic Scholar integration) (Kinney et al., 2023; Semantic Scholar, 2024); (ii) Neo4j-backed storage and Cypher querying (Neo4j Inc., 2024); (iii) interactive graph and semantic visualizations (Cytoscape.js and D3.js) (Bostock et al., 2011; Franz et al., 2023); and (iv) an AI "Talk To Your Data" interface that supports both simple retrieve-and-generate and ReAct-style agentic workflows (Yao et al., 2022).

Architecture. The system uses a containerized, microservices design orchestrated with Docker Compose (Docker, Inc., 2024; Merkel, 2014). Primary services include: a FastAPI service for HTTP coordination (FastAPI Contributors, 2024), a Neo4j database for graph storage (Neo4j Inc., 2024), and a Knowledge Graph service for document processing/embeddings/AI tasks. Services communicate asynchronously via RabbitMQ, enabling horizontal scaling (VMware, Inc., 2024).

MCP integration. Nexarag ships an MCP-compatible server that exposes graph querying, semantic search over embedded content, and external search via Semantic Scholar to any MCP-enabled LLM (local via Ollama or remote via hosted providers) (Anthropic, 2024; Model Context Protocol Contributors, 2024; Ollama Team, 2024; OpenAI, 2023). This standardizes context delivery and promotes reproducible prompt-driven research workflows.

Install & minimal run. (see repository docs for full instructions)

```
# CPU example
docker compose -f docker-compose.cpu.yml up -d
# or on macOS
docker compose -f docker-compose.macos.yml up -d
```

Optionally pull local models for embedding/LLM integration with Ollama (Ollama Team, 2024); for example, a long-context text embedder like Nomic Embed (Nussbaum et al., 2025):

```
# inside the Ollama container or on macOS host
ollama pull nomic-embed-text:v1.5
ollama pull gemma3:1b
```

Repository: https://github.com/REPLACE-WITH-YOUR-ORG/nexarag

License: GNU General Public License v3.0.

Quality control

Nexarag emphasizes verifiable operation through containerized deployment and a guided quick start (Boettiger, 2015). Reviewers can launch the full stack with Docker Compose, query/persist KGs in Neo4j, and exercise end-to-end flows (semantic search, citation expansion, MCP tools). A worked MCP chat transcript and an automatically generated literature review illustrate that the system's graph building, retrieval, and reporting features execute as described. The repository includes example datasets/notebooks and scripts for running tests where applicable, supporting broader reproducibility goals in research practice (Rothacher et al., 2023).

Use cases

- **Reproducible literature reviews.** Build a KG from a seed set (e.g., via BibTeX), expand by citations, and generate a structured review through the MCP interface (Sahlab et al., 2022).
- **Private research contexts.** Run entirely offline (air-gapped) with local LLMs for sensitive domains (e.g., healthcare, legal, proprietary research) (Boettiger, 2015).

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- **Collaborative curation.** Share/export/import graphs across teams to support longitudinal projects.

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References

Please place bibliographic entries for works discussed in a `paper.bib` file and cite them inline (e.g., `@key`). For submission, ensure all citation keys resolve and that venues are written out in full.

Anthropic. (2024). *Introducing the model context protocol*. https://www.anthropic.com/news/model-context-protocol. https://www.anthropic.com/news/model-context-protocol

Boettiger, C. (2015). An introduction to docker for reproducible research. *ACM SIGOPS Operating Systems Review*, *49*(1), 71–79. https://doi.org/10.1145/2723872.2723882

Bostock, M., Ogievetsky, V., & Heer, J. (2011). D3: Data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, *17*(12), 2301–2309. https://doi.org/10.1109/TVCG.2011.185

Docker, Inc. (2024). *Docker: Accelerated container application development*. https://www.docker.com/. https://www.docker.com/

FastAPI Contributors. (2024). *FastAPI: Modern, fast (high-performance), web framework for building APIs with python 3.6+ based on standard python type hints*. https://fastapi.tiangolo.com/. https://fastapi.tiangolo.com/

Franz, M., Lopes, C. T., Fong, D., Kucera, M., Cheung, M., Siper, M. C., Huck, G., Dong, Y., Sumer, O., & Bader, G. D. (2023). Cytoscape.js 2023 update: A graph theory library for visualization and analysis. *Bioinformatics*, *39*(1), btad031. https://doi.org/10.1093/bioinformatics/btad031

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H., & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey. *arXiv Preprint arXiv:2312.10997*, *2*.

Guu, K., Lee, K., Tung, Z., Pasupat, P., & Chang, M.-W. (2020). REALM: Retrieval-augmented language model pre-training. *Proceedings of the 37th International Conference on Machine Learning*.

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., & Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Trans. Inf. Syst.*, *43*(2). https://doi.org/10.1145/3703155

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Comput. Surv.*, *55*(12). https://doi.org/10.1145/3571730

Kinney, R. M., Anastasiades, C., Authur, R., Beltagy, I., Bragg, J., Buraczynski, A., Cachola, I., Candra, S., Chandrasekhar, Y., Cohan, A., Crawford, M., Downey, D., Dunkelberger, J., Etzioni, O., Evans, R., Feldman, S., Gorney, J., Graham, D. W., Hu, F. Q., … Weld, D. S. (2023). The semantic scholar open data platform. *ArXiv*, *abs/2301.10140*.

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https://api.semanticscholar.org/CorpusID:256194545

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Proceedings of the 34th International Conference on Neural Information Processing Systems*.

Merkel, D. (2014). Docker: Lightweight linux containers for consistent development and deployment. *Linux Journal*, *2014*(239), 2.

Model Context Protocol Contributors. (2024). *Model context protocol: A protocol for seamless integration between LLM applications and external data sources*. https://github.com/modelcontextprotocol. https://github.com/modelcontextprotocol

Neo4j Inc. (2024). *Neo4j graph database platform*. https://neo4j.com/product/neo4j-graph-database/. https://neo4j.com/product/neo4j-graph-database/

Nussbaum, Z., Morris, J. X., Mulyar, A., & Duderstadt, B. (2025). Nomic embed: Training a reproducible long context text embedder. *Transactions on Machine Learning Research*. https://openreview.net/forum?id=IPmzyQSiQE

Ollama Team. (2024). *Ollama: Get up and running with llama 3.2, mistral, gemma 2, and other large language models*. https://ollama.com/. https://ollama.com/

OpenAI. (2023). *OpenAI API*.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 3973–3983. http://arxiv.org/abs/1908.10084

Reinanda, R., Meij, E., & Rijke, M. de. (2020). Knowledge graphs: An information retrieval perspective. *Foundations and Trends in Information Retrieval*, *14*(2-3), 289–444. https://doi.org/10.1561/1500000063

Rothacher, L., Engel, C., Garbeva, P., Grüning, B., Goble, C., Gundersen, S., Ioannidis, J. P., Marwick, B., Parsons, S., Pronk, T. E., & others. (2023). Eleven strategies for making reproducible research and open science training the norm at research institutions. *eLife*, *12*, RP89736. https://doi.org/10.7554/eLife.89736

Sahlab, N., Kahoul, H., Jazdi, N., & Weyrich, M. (2022). A knowledge graph-based method for automating systematic literature reviews. *arXiv Preprint arXiv:2208.02334*. https://doi.org/10.48550/arXiv.2208.02334

Semantic Scholar. (2024). *Semantic scholar academic graph API*. https://www.semanticscholar.org/product/api. https://www.semanticscholar.org/product/api

VMware, Inc. (2024). *RabbitMQ: The most widely deployed open source message broker*. https://www.rabbitmq.com/. https://www.rabbitmq.com/

Wang, M., Chen, L., Cheng, F., Liao, S., Zhang, X., Wu, B., Yu, H., Xu, N., Zhang, L., Luo, R., Li, Y., Yang, M., Huang, F., & Li, Y. (2024). Leave no document behind: Benchmarking long-context LLMs with extended multi-doc QA. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 5627–5646. https://doi.org/10.18653/v1/2024.emnlp-main.322

Xu, Z., Cruz, M. J., Guevara, M., Wang, T., Deshpande, M., Wang, X., & Li, Z. (2024). Retrieval-augmented generation with knowledge graphs for customer service question answering. *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2905–2909. https://doi.org/10.1145/3626772.3661370

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2022). ReAct: Synergizing reasoning and acting in language models. *arXiv Preprint arXiv:2210.03629*.

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https://doi.org/10.48550/arXiv.2210.03629

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